

Utkarsh Mishra

Analytics Engineer | Decision Intelligence & Data Systems
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IMPACT HIGHLIGHTS

- Reduced decision turnaround time by 25–30% by designing AI-driven dashboards and predictive analytics systems across multi-industry client engagements.
- Improved marketing conversion performance by 17% through regression-based budget optimization and scenario modeling frameworks.
- Increased sales productivity by 30% by leading CRM system implementation and reporting modernization initiatives.
- Generated ~\$25K in annual cost savings through workflow optimization and data-driven process redesign.

SUMMARY

Analytics Engineer with 5+ years of experience building scalable data models, automated reporting systems, and AI-enabled analytics solutions. Strong foundation in advanced SQL, dimensional modeling, KPI governance, and cloud data platforms. Focused on designing clean data architectures and layering intelligent automation to support faster, more reliable business decisions.

TECHNICAL SKILLS

- Programming & Querying: SQL (CTEs, Window Functions, Query Optimization), Python (Pandas, NumPy, Scikit-Learn), Git
- Analytics Engineering & Data Modeling: Dimensional Modeling (Star Schema), Data Warehousing, ELT/ETL Pipelines, Data Transformation, Data Validation, KPI Governance
- Cloud & Platforms: Snowflake, Azure Synapse, Azure Data Factory, Databricks, BigQuery, AWS EC2
- Business Intelligence: Power BI (DAX, RLS, Dataflows), Tableau, Executive Dashboarding
- AI & Advanced Analytics: Time-Series Forecasting (Prophet), XGBoost, Regression Modeling, SHAP, NLP (BERT), RAG Architecture
- Infrastructure & Deployment: Docker, Multi-Service API Deployment, Uvicorn

PROFESSIONAL EXPERIENCE

Data Strategy Intern | Cognitive Sprints AI , Dallas, TX [Aug 2025 – Present]

- Executing end-to-end analytics and AI initiatives across manufacturing, healthcare, and IT clients, translating ambiguous business requirements into scalable data models, KPI architectures, and decision automation layers.
- Designed and deployed executive dashboards and predictive models (SQL, Python, Power BI), reducing decision turnaround time by 25–30%.
- Built and automated ELT pipelines and reporting systems, improving data reliability, governance, and metric standardization across stakeholder teams.
- Contributed to AI security testing and orchestration efforts, including RAG evaluation, prompt injection red-teaming, and blue-team validation workflows to assess model robustness and access control integrity.
- Supported deployment of containerized AI services across sandbox and GPU environments, participating in inference routing, model orchestration, and retrieval performance evaluation.

Projects IT Intern | University of Texas System, Richardson, TX, USA [June 2024 - Aug 2024]

- Architected and deployed 5+ enterprise Power BI dashboards analyzing 50K+ operational records, strengthening KPI governance and reporting consistency across departments.
- Established real-time performance monitoring workflows enabling leadership to proactively address operational bottlenecks and optimize resource allocation.
- Developed predictive threshold forecasting models to flag deficit work-hour risk trends, supporting early intervention and capacity planning decisions.

Business Data Analyst | Webcraft IT, Indore, India [July 2019 - July 2023]

- Designed and scaled executive analytics dashboards (Power BI, Tableau) translating operational and customer datasets into structured decision frameworks for leadership.
- Led cross-functional analytics initiatives across five teams, standardizing reporting workflows and reducing project delivery delays by 12.5%.
- Directed end-to-end CRM implementation (requirements, configuration, reporting architecture), increasing sales productivity by 30% through improved pipeline visibility.
- Identified workflow inefficiencies using data-driven process analysis, delivering ~\$25K in annual cost savings and reducing customer complaints by 20%.
- Facilitated stakeholder workshops to translate business objectives into system-level data specifications for digital transformation initiatives.

Financial Analyst Intern | Tata Consultancy Services (TCS) Mumbai, India [May 2017 – Jun 2017]

- Analyzed financial performance across 20+ corporate entities, evaluating revenue trends and cost structures to support senior stakeholder reporting.
- Utilized Oracle GL and enterprise reporting tools to extract, reconcile, and validate financial datasets.

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SELECTED SYSTEMS & DECISION ARCHITECTURES

DASHAI - AI-Powered BI Model | MSBA-AI, UTD (Python • Streamlit • XGBoost • SHAP • Prophet • NLP • Power BI)

Designed and built a modular analytics platform integrating predictive modeling, clustering, NLP-driven text analysis, and model explainability into a unified decision layer.

- Architected end-to-end workflow combining data ingestion, feature engineering, forecasting (Prophet/XGBoost), and SHAP-based explainability.
- Automated insight generation across financial, operational, and marketing datasets, reducing manual analysis cycles by 30%.
- Developed executive-facing dashboards translating model outputs into scenario-driven decision workflows.
- Structured system for extensibility across forecasting, recommendation logic, and segmentation pipelines.

AMIREAL - Authenticity Ledger & Verification Platform (Python • Cryptographic Hashing • Metadata Validation • Streamlit)

Designed a tamper-evident verification system to establish trust and auditability for digital media assets.

- Architected cryptographic hashing framework generating immutable digital fingerprints for file-level integrity validation.
- Built metadata extraction and validation pipeline to detect structural inconsistencies and enforce integrity checks.
- Developed verification interface enabling hash comparison and authenticity validation workflows.
- Structured architecture for future integration of AI-based synthetic media detection and governance controls.

ELECKART – Marketing Decision Optimization Engine (Python • SQL • Regression Modeling • Scenario Simulation • Power BI)

Built a regression-based marketing attribution and budget optimization framework designed to convert raw campaign data into actionable investment decisions.

- Engineered multi-channel attribution model analyzing 12 months of campaign performance data.
- Identified reallocation strategy improving conversion performance by 17%.
- Designed scenario simulation dashboards enabling forward-looking ROI projections (~12% uplift potential).
- Translated statistical outputs into budget allocation recommendations aligned with executive KPIs.

AI SECURITY & INFRASTRUCTURE SYSTEMS

- Evaluated retrieval-augmented generation (RAG) systems for prompt injection vulnerabilities and access-control boundary enforcement.
- Implemented dual-instance AWS architecture separating GPU-based inference from application services to improve isolation and scalability.
- Deployed containerized AI services using Docker across sandboxed environments to support controlled testing and model evaluation.

ADDITIONAL ANALYTICS SYSTEMS

FIFA Sentiment Analysis (BERT) | Market Basket Analysis | Inventory Forecasting | Real-Time Streaming Pipelines | GeminiLLM-NLP | AI Referee Evaluation System | CRM Analytics & Reporting Modernization | Keyword Gap & SEO Performance Analytics | Data Center Cost Modeling (TCO Optimization) | Healthcare Distribution Analytics

EDUCATION

MS in Business Analytics and Artificial Intelligence

University of Texas at Dallas, TX (USA)

[Aug 2023 - May 2025]

NLP, Applied ML, AI in Marketing, Econometrics and Time Series, Predictive Modeling, Prescriptive Modeling, Web Analytics, DBMS, Visualization, Statistics

Executive PG in Data Science

International Institute of Information Technology (India)

[May 2020 - May 2021]

Python (Bootcamp), Data Toolkit, Data Visualization, Advanced Machine Learning, Business Requirements for Data Science, Capstone

Bachelor of Business Management (Marketing Honors)

Xavier Institute of Management- Bhubaneswar (India)

[Apr 2016 - Apr 2019]

Business Communication, Business Management, Sustainability, Statistics, Marketing, Digital Marketing, Sales, Supply Chain, Business Finance, Accounting